1.Project Overview

Target Corporation, one of the largest retail chains in the United States, operates through hypermarkets, discount department stores, and e-commerce platforms. With growing competition from Walmart, Amazon, and Costco, retaining loyal customers has become a crucial challenge. Target collects extensive data from customer transactions, loyalty memberships, and online interactions, but its current reporting lacks analytical depth to:

● Understand why customers are churning.

● Identify loyal vs. at-risk customers.

● Measure the impact of loyalty tiers, promotions, and influencer-driven campaigns.

● Guide region- and channel-specific retention strategies.

This report presents a comprehensive, data-driven analysis of Target's customer retention, loyalty program engagement, promotion efficiency, and store/channel performance. Utilizing a dataset representing 200 distinct customer profiles and 1,000 transaction records, we engineered a relational Star Schema model in Power BI Desktop to isolate critical friction points and map long-term customer value.

2. Project Objectives

The core objectives of this data modeling and business intelligence initiative are to:

- **Consolidate Fragmented Data Silos**: Ingest and connect customer demographics, purchase history, store locations, transactional data, and loyalty program details into a single, cohesive reporting model.
- **Enable Dynamic Cohort Segmentation**: Deliver robust segmentation matrices across high-value (High-CLV), repeat, and churned customer groups to monitor active and at-risk segments in real-time.
- **Strengthen Engagement and Retention**: Uncover specific performance variations across regions, sales channels, store formats, and promotional

programs to form actionable, data-backed operational recommendations that reduce churn and enhance high-tier loyalty.

3. Core Business KPIs

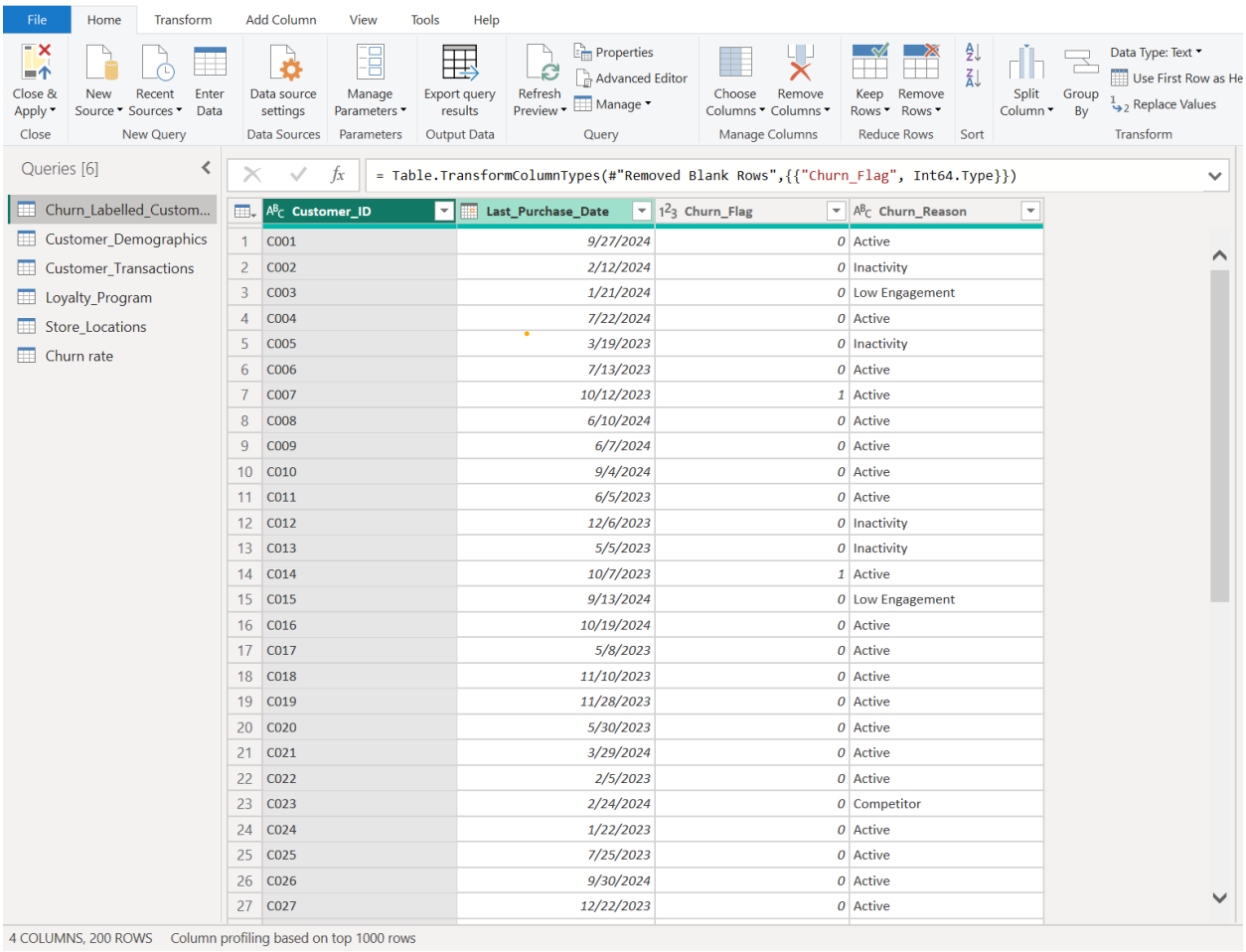
- **Total Customer Base**: 200 Unique Customers
- **Retained (Repeat) Customers**: 151 Active Shoppers (75.50%)
- **Churned Customers**: 49 Terminated Accounts (24.50%)
- **Target Baseline Retention Rate**: 75.50%
- **Target Baseline Churn Rate**: 24.50%
- **Average Customer Lifetime Value (CLV)**: 509.35 annually

TASK TO BE PERFORMED:

TASK 1: Data Modeling & Cleaning Task

1.1: Load and transform datasets in Power Query

The five source files (Customer_Demographics.csv, Customer_Transactions.csv, Store_Locations.csv, Loyalty_Program.csv, Churn_Labelled_Customers.csv) were imported into the Power Query Editor in Power BI Desktop.

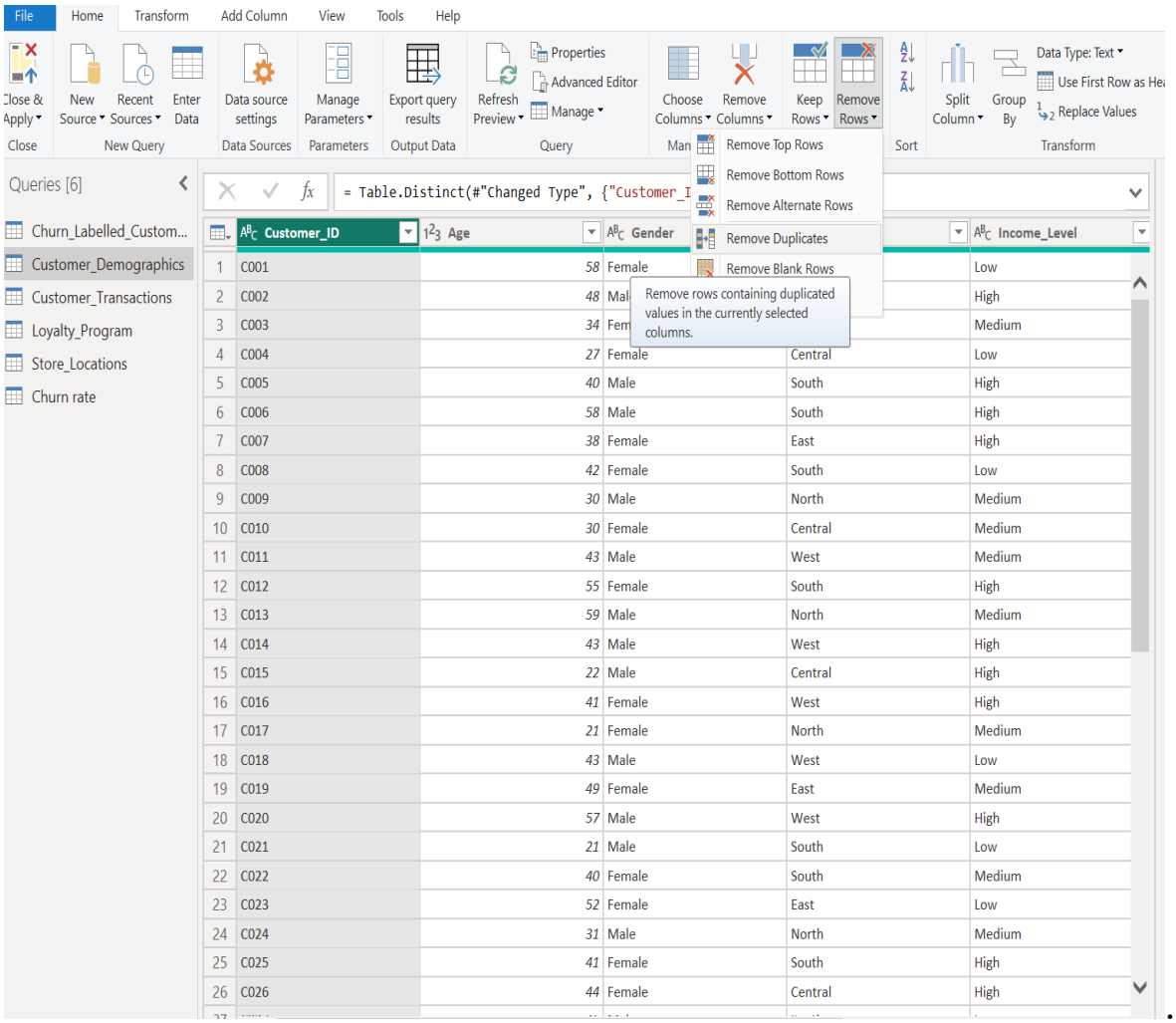


	Customer_ID	Last_Purchase_Date	Churn_Flag	Churn_Reason
1	C001	9/27/2024	0	Active
2	C002	2/12/2024	0	Inactivity
3	C003	1/21/2024	0	Low Engagement
4	C004	7/22/2024	0	Active
5	C005	3/19/2023	0	Inactivity
6	C006	7/13/2023	0	Active
7	C007	10/12/2023	1	Active
8	C008	6/10/2024	0	Active
9	C009	6/7/2024	0	Active
10	C010	9/4/2024	0	Active
11	C011	6/5/2023	0	Active
12	C012	12/6/2023	0	Inactivity
13	C013	5/5/2023	0	Inactivity
14	C014	10/7/2023	1	Active
15	C015	9/13/2024	0	Low Engagement
16	C016	10/19/2024	0	Active
17	C017	5/8/2023	0	Active
18	C018	11/10/2023	0	Active
19	C019	11/28/2023	0	Active
20	C020	5/30/2023	0	Active
21	C021	3/29/2024	0	Active
22	C022	2/5/2023	0	Active
23	C023	2/24/2024	0	Competitor
24	C024	1/22/2023	0	Active
25	C025	7/25/2023	0	Active
26	C026	9/30/2024	0	Active
27	C027	12/22/2023	0	Active

1.2 Handle duplicates, missing values, and ensure correct data types

The following transformations were executed:

- **Deduplication**: Enforced table integrity by applying **Remove Duplicates** to the unique primary keys (Customer_ID and Store_ID).
- **Null Handling**: Cleaned missing fields across the dataset using **Remove Blank Rows** and replaced empty text fields.
- **Explicit Data Typing**: Standardized all columns into strict data types

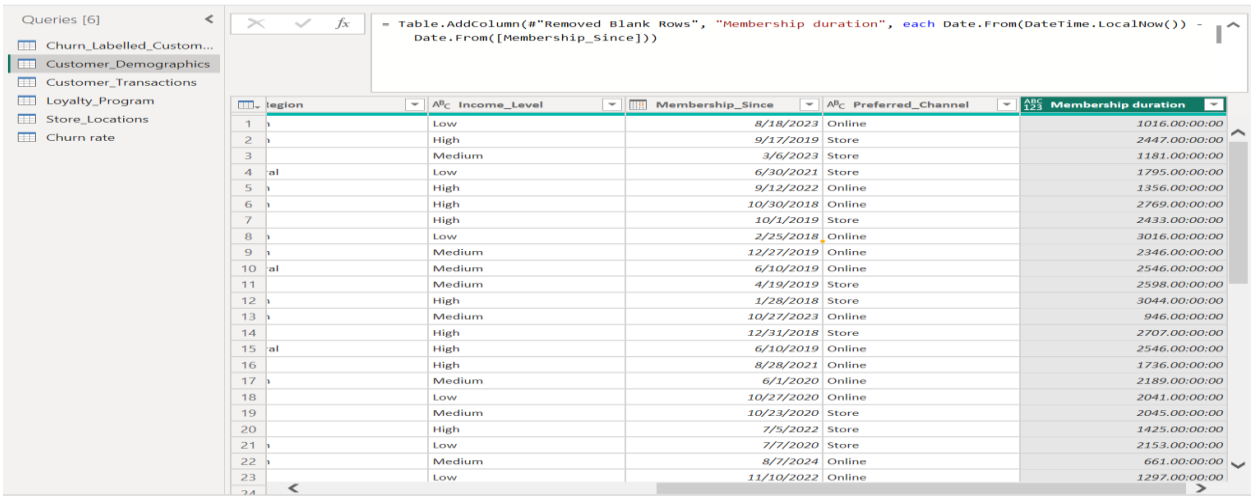


Task 1.3: Create calculated columns

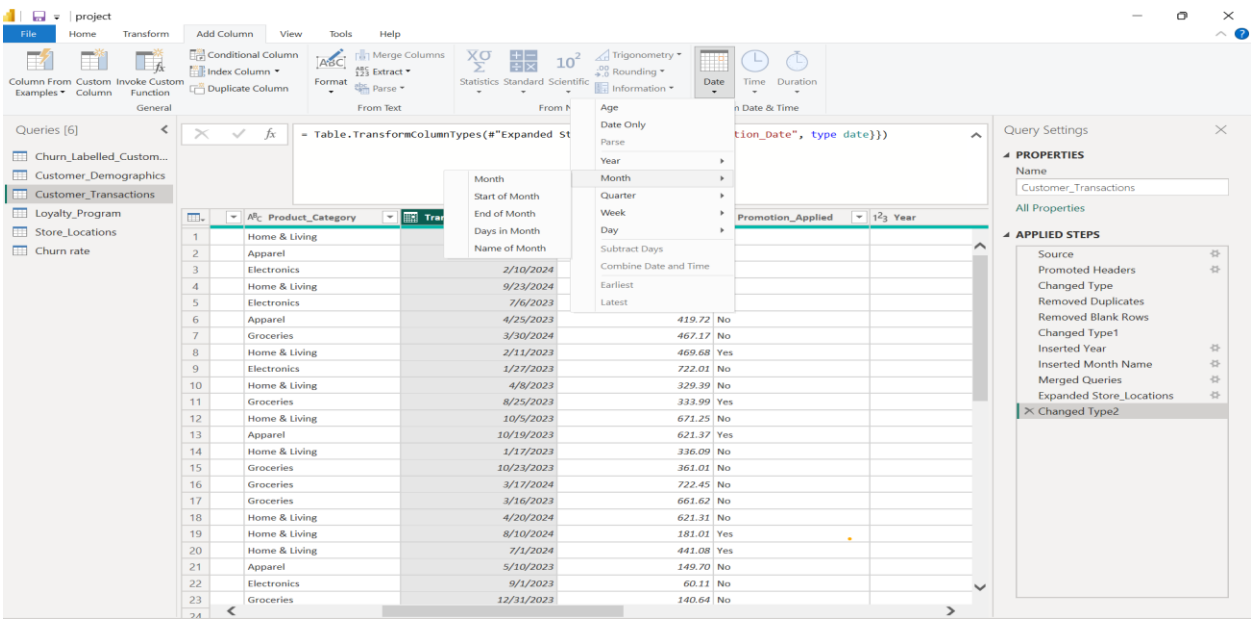
A) Membership_Duration= Today – Membership_Since

Step 1:Select Customer_Demographics and go to Add Column click the custom column to enter formula.

Duration.Days(Date.From(DateTime.LocalNow()) - Date.From([Membership_Since]))

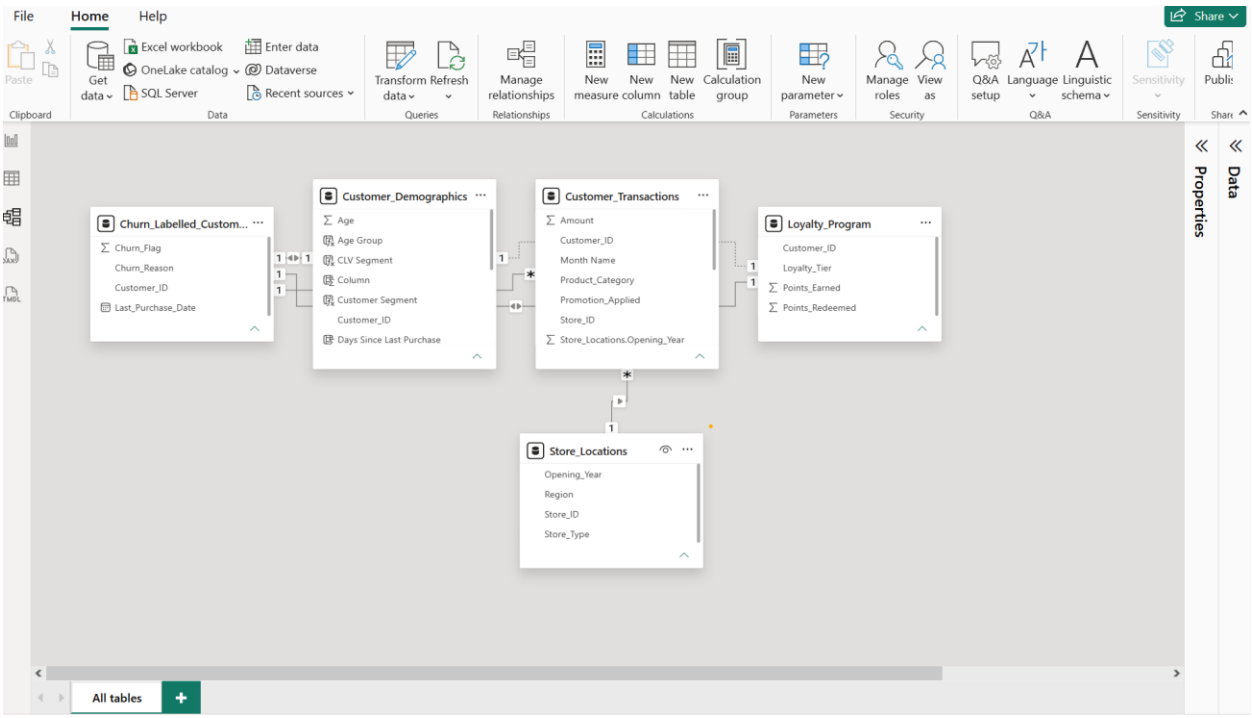


B) Extract Transaction_Year, Transaction_Month Step1:Go to: Customer_Transactions table and Select Transaction_Date column and Add Column → Date → Year → Year



Task 1.4: Create a basic Data Model view

To organize data relationship directions and prevent circular reference issues, we set up a Star Schema inside the Power BI Model View.



Task 2: Churn & Retention Metrics

TASK 2.1: Create Churn Rate KPI

Using Dax to create the churn rate%

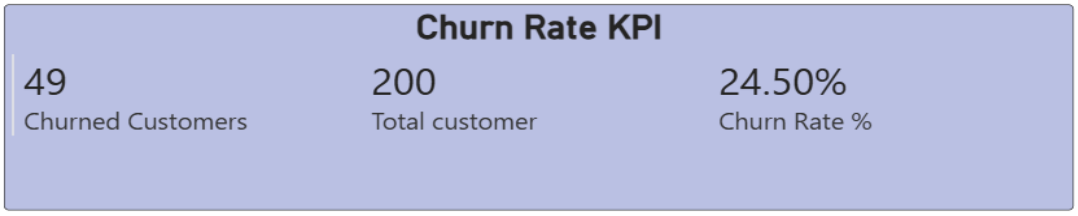
FORMULA:

TotalCustomers=DISTINCTCOUNT(Customer_Demographics

[Customer_ID])

ChurnedCustomers=CALCULATE(DISTINCTCOUNT(Churn_Labelled_Customers[Customer_ID]),Churn_Labelled_Customers[Churn_Flag=1)

Churn Rate % = DIVIDE([Churned Customers], [Total Customers],0)



Task 2.2: Visualize churn rate by

A) Churn rate% by Region

Insert → Bar Chart → Clustered Bar Chart.

Y-Axis: Customer_Demographics[Region].

X-Axis: [Churn_Rate (%)] measure

Format → Title: 'Churn Rate by Region'

B) Churn rate% by Income Group

Insert → Donut Chart.

Legend: Customer_Demographics[Income_Level].

Values: [Churned_Customers]

C) Churn rate% by Channel (Store/Online)

Insert → Clustered Column Chart.

X-Axis: Customer_Demographics[Preferred_Channel].

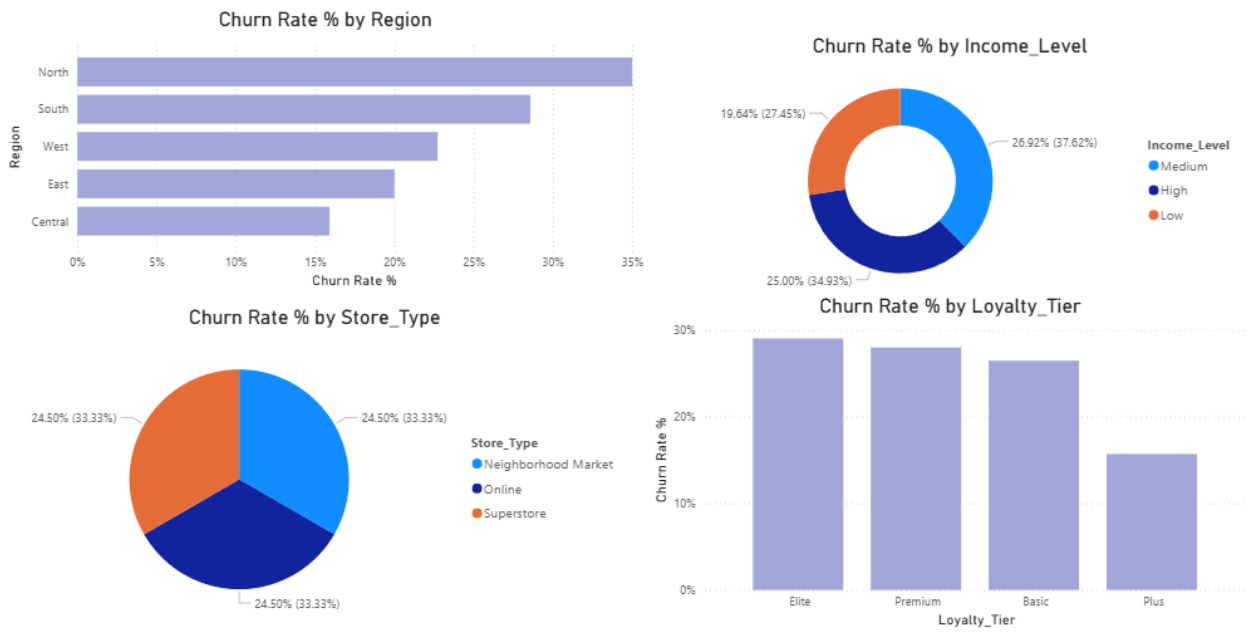
Y-Axis: [Churn_Rate (%)].

D) Churn rate% by Loyalty Tier

Insert → Clustered Bar Chart.

Y-Axis: Loyalty_Program[Loyalty_Tier].

X-Axis: [Churn_Rate (%)].



Task 2.3: Funnel Chart : Total Customers → Repeat Customers → Churned

Step 1: Create Repeat Customers Measure

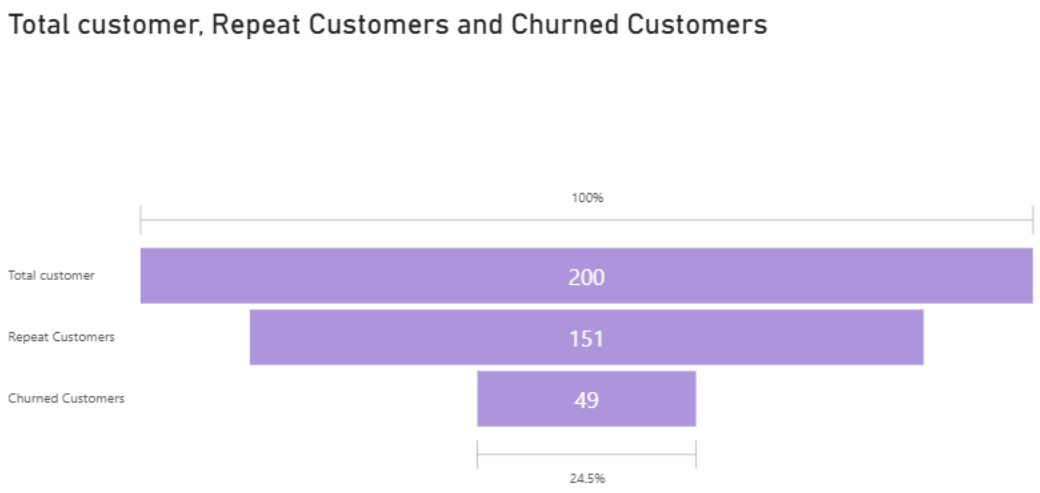
Formula: Repeat Rate % = DIVIDE([Total Customers] - [Churned Customers], [Total Customers], 0) * 100

Step 2:

Insert → Funnel Chart.

Category: use a manual table or calculated column with values: 'Total Customers', 'Repeat Customers', 'Churned Customers'.

Values: [Total_Customers], [Repeat_Customers], [Churned_Customers].



Task 3.1: Segment customers:

○ Low-Tier: 0–3 purchases ○ Mid-Tier: 4–8 purchases ○ High-Tier: 9+ purchases

Step 1: Create new column to purchase count

```
Purchase Count =
VAR CurrentCustomerID = Customer_Demographics[Customer_ID]
RETURN
COUNTROWS(
   FILTER(
      Customer_Transactions,
      Customer_Transactions[Customer_ID] = CurrentCustomerID ))
```

Step 2: Create customer segment to new column

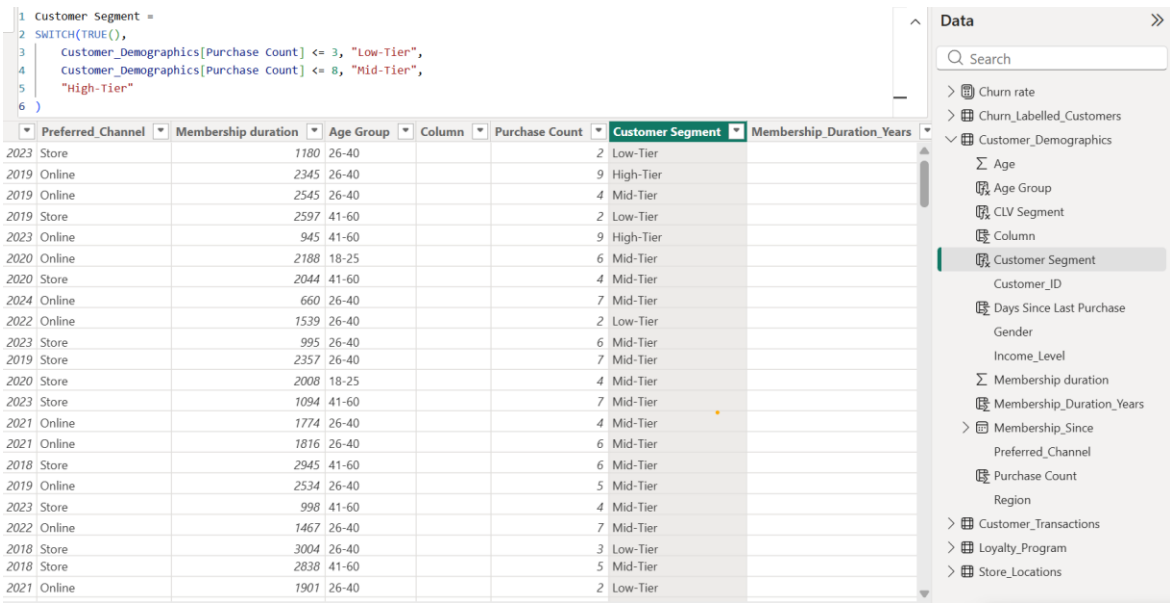
Customer Segment =

SWITCH(TRUE(),

Customer_Demographics[Purchase Count] <= 3, "Low-Tier",

Customer_Demographics[Purchase Count] <= 8, "Mid-Tier",

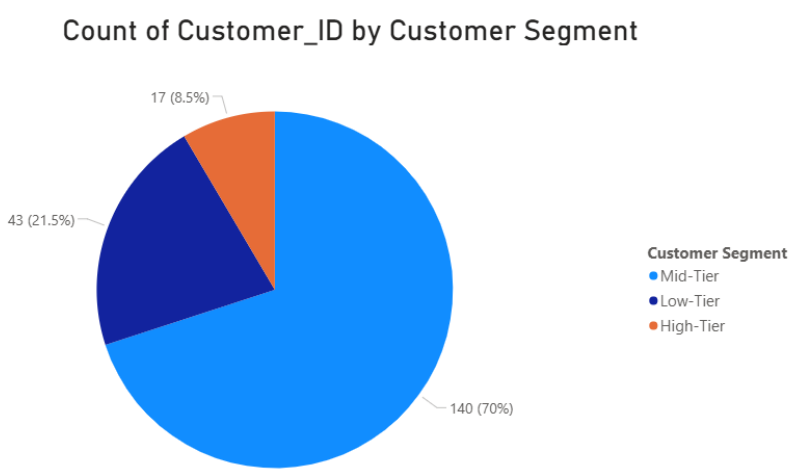
"High-Tier")



```
Customer Segment =
SWITCH(TRUE(),
    Customer_Demographics[Purchase Count] <= 3, "Low-Tier",
    Customer_Demographics[Purchase Count] <= 8, "Mid-Tier",
    "High-Tier"
)
```

	Preferred_Channel	Membership duration	Age Group	Column	Purchase Count	Customer Segment	Membership_Duration_Years
2023	Store	1180	26-40		2	Low-Tier	
2019	Online	2345	26-40		9	High-Tier	
2019	Online	2545	26-40		4	Mid-Tier	
2019	Store	2597	41-60		2	Low-Tier	
2023	Online	945	41-60		9	High-Tier	
2020	Online	2188	18-25		6	Mid-Tier	
2020	Store	2044	41-60		4	Mid-Tier	
2024	Online	660	26-40		7	Mid-Tier	
2022	Online	1539	26-40		2	Low-Tier	
2023	Store	995	26-40		6	Mid-Tier	
2019	Store	2357	26-40		7	Mid-Tier	
2020	Store	2008	18-25		4	Mid-Tier	
2023	Store	1094	41-60		7	Mid-Tier	
2021	Online	1774	26-40		4	Mid-Tier	
2021	Online	1816	26-40		6	Mid-Tier	
2018	Store	2945	41-60		6	Mid-Tier	
2019	Online	2534	26-40		5	Mid-Tier	
2023	Store	998	41-60		4	Mid-Tier	
2022	Online	1467	26-40		7	Mid-Tier	
2018	Store	3004	26-40		3	Low-Tier	
2018	Store	2838	41-60		5	Mid-Tier	
2021	Online	1901	26-40		2	Low-Tier	

Visualize Customer ID by Customer segment:



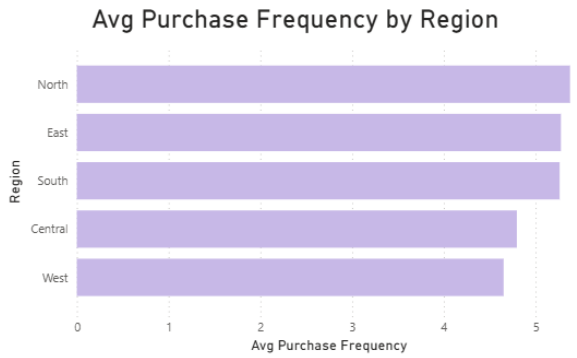
Task 3.2: Compare avg. purchase frequency by Region, Age Group, Loyalty Tier

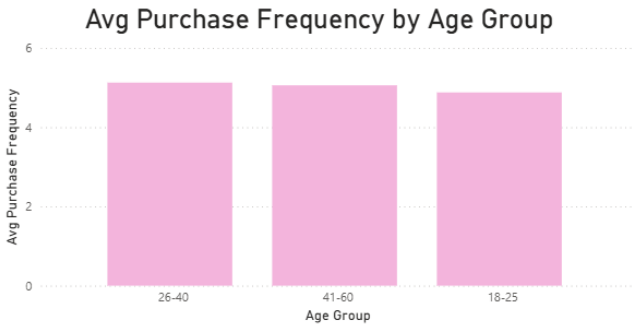
Visualize avg. purchase frequency by Region

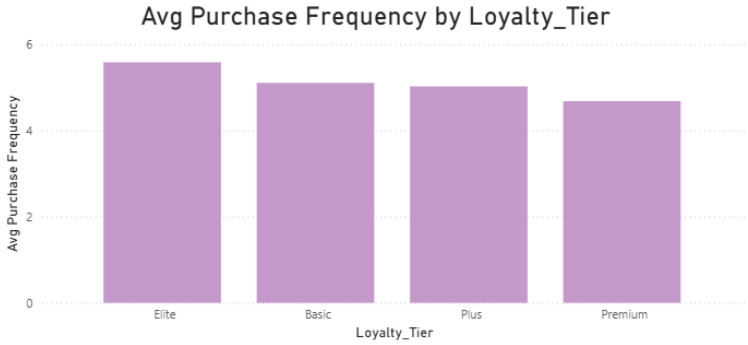
Visual: Clustered bar Chart Field: Region->axis , avg. purchase frequency-> values Visualize avg. purchase frequency by Age Group

Visual: Clustered column Chart Field: AgeGroup,->axis, avg. purchase frequency-> values Visualize avg. purchase frequency by Loyalty Tier

Visual: Clustered bar Chart Field: Loyalty Tier ->axis , avg. purchase frequency-> values

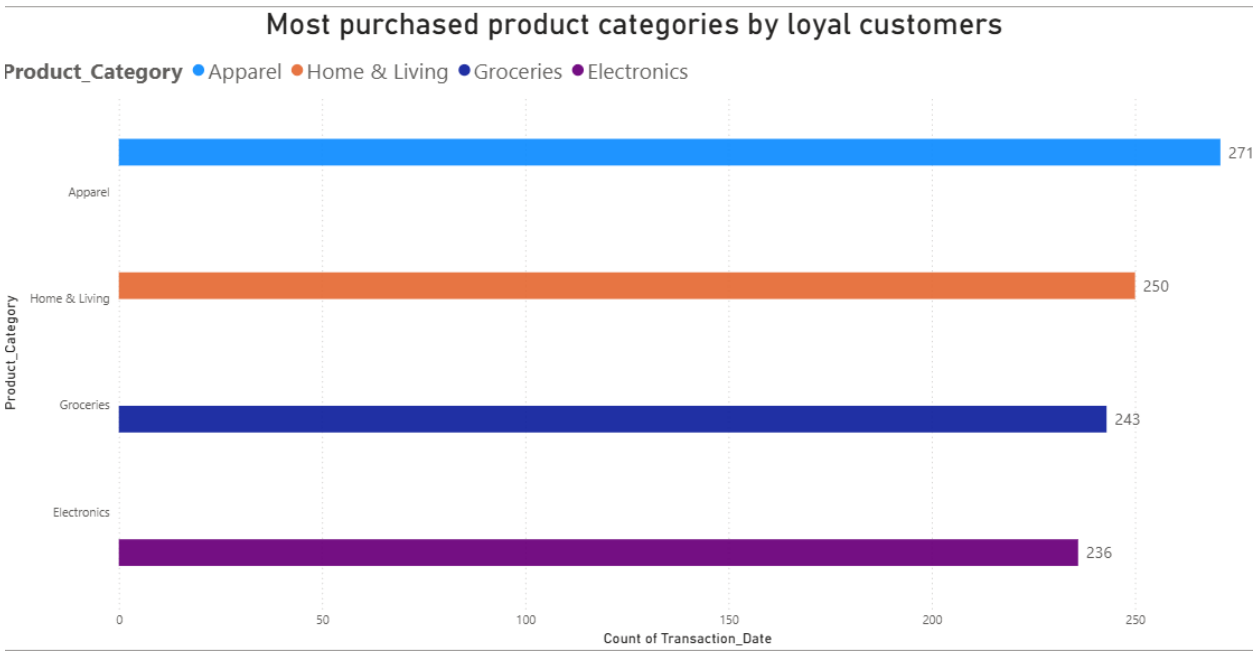






Task3.3: Identify most purchased product categories by loyal customers

Visualize transaction_ID by Product_Category Visual: Clustered bar Chart Field: Product_Category ->axis , transaction id-> values



Task4: Promotion & Loyalty Impact

Task4.1: Calculate % of transactions with promotion applied.

Step1: create total transaction measure

Total Transactions = COUNT(Customer_Transactions[Transaction_ID])

Step 2: create promotional transaction measure with applied “yes” in promotional applied

Promotional Transactions = CALCULATE(
COUNT(Customer_Transactions[Transaction_ID]),
Customer_Transactions[Promotion_Applied] = "Yes")

Step3: create promotion percentage measure

Promotion % = FORMAT(DIVIDE([Promotional Transactions], [Total

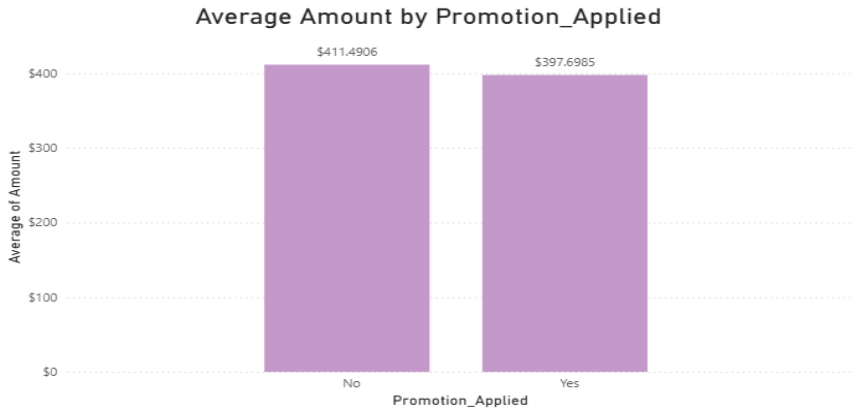
Transactions], 0), "0.00%")

Calculate % of transactions with promotion applied. Calculate % of transactions with promotion applied.

Task 4.2: Compare average purchase amount with vs. without promotions.

Visualize average purchase amount with vs. without promotions.

Visual: Clustered Column Chart Field: Promotion applied->axis , average purchase amount -> values



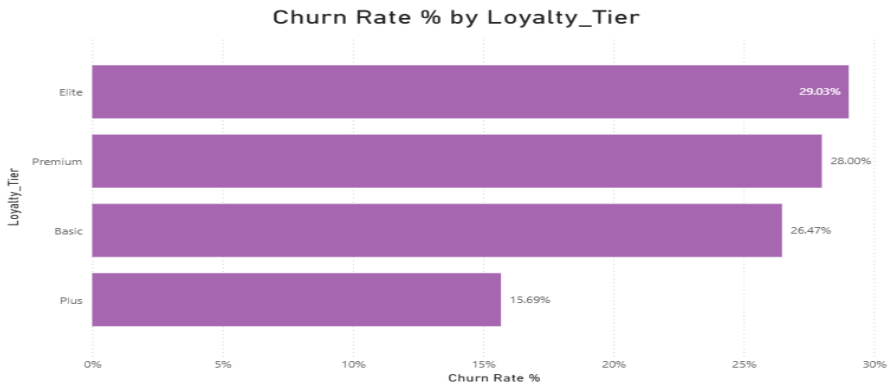
This reveals that promotions are actually diluting Target's average basket sizes by $13 per transaction. This suggests that discounts are primarily being applied to items customers would have purchased anyway, rather than successfully encouraging shoppers to add extra, high-value items to their carts.

Task 4.3: Analyze churn rate across loyalty tiers.

Visualize churn rate %by loyalty tiers

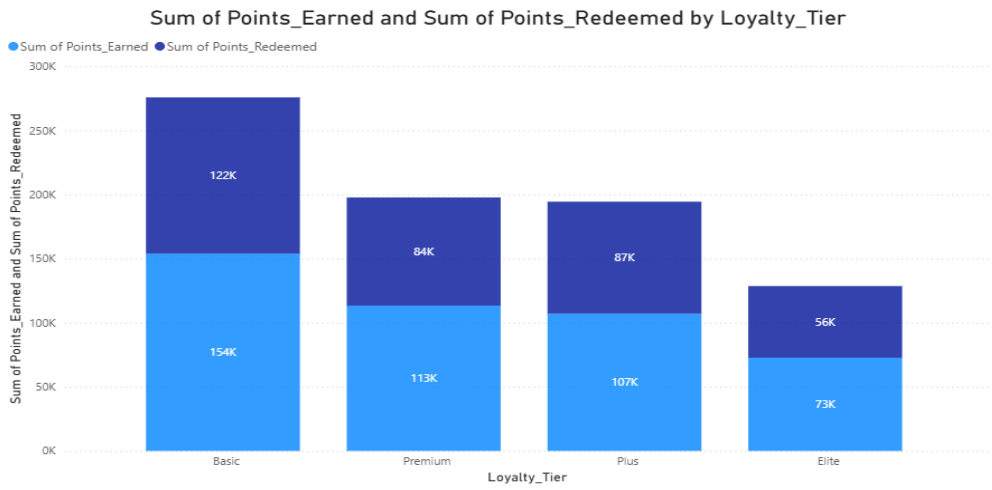
Visual: Clustered bar Chart Field: Loyalty Tier ->axis ,

churn rate%->values



TASK 4.4: Visualize Points Earned vs. Redeemed by Tier

Insert a Clustered Column Chart. Place Loyalty_Program[Loyalty_Tier] on the X-Axis. Add two Y-Axis series:
SUM(Loyalty_Program[Points_Earned]) labelled 'Points Earned' and
SUM(Loyalty_Program[Points_Redeemed]) labelled 'Points Redeemed'



TASK 4.5: Give recommendations to improve redemption & retention.

- Target Mid-Tier customers with personalised promotions; avoid blanket discounting for Elite tier which buys without incentives.
- Launch a time-limited 'Redeem Now' campaign with bonus value redemption events to improve programme engagement
- Introduce a 'Fast-Track to Plus' incentive: double points for 60 days to accelerate Basic → Plus conversion and reduce churn risk.

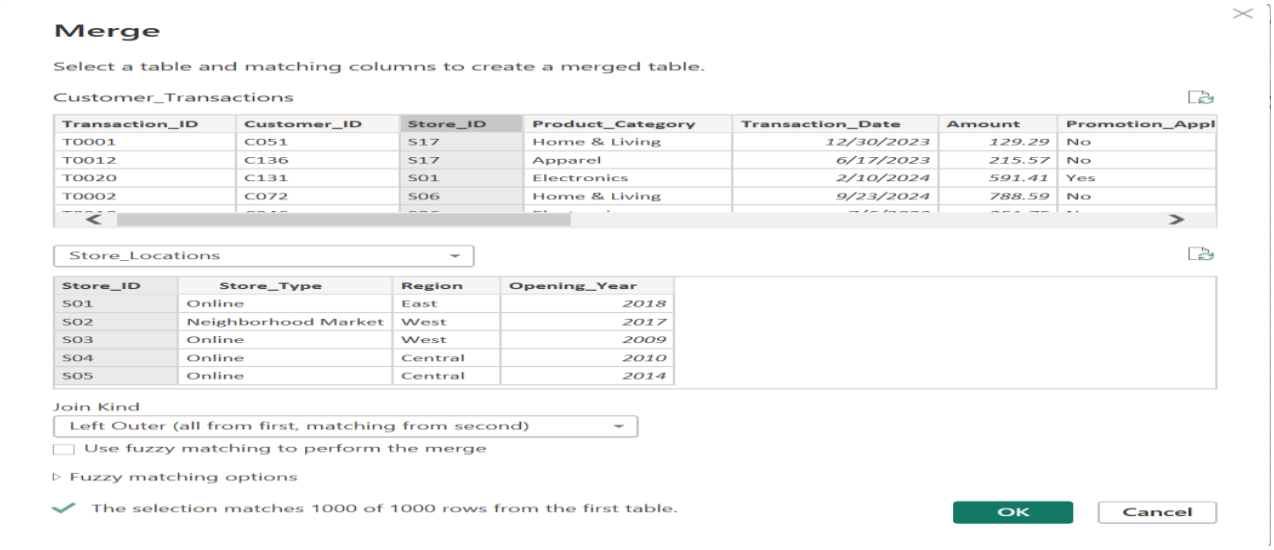
TASK 5: Store & Channel Performance vs Retention

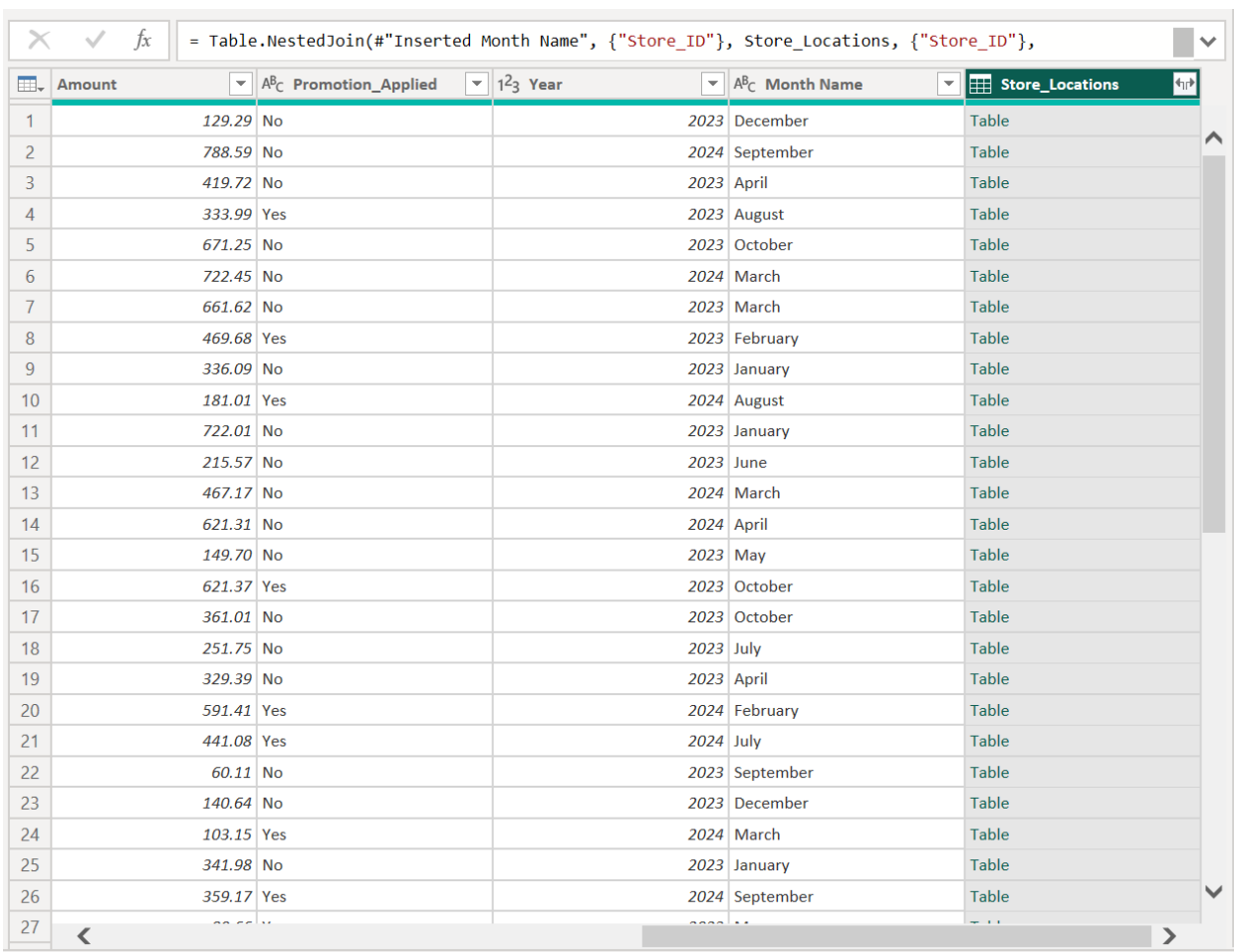
Task 5.1: Merge store data with transactions

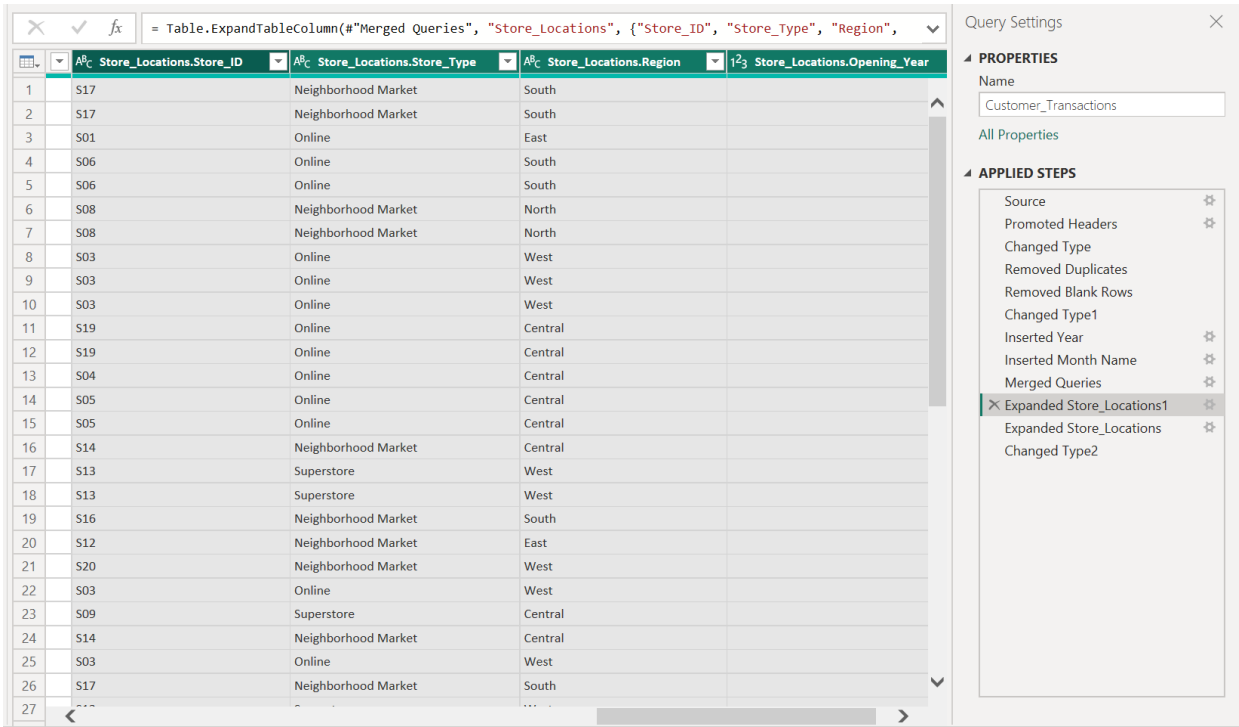
Step 1 : In Power Query, open the Customer_Transactions query.

Step 2 : Click Home → Merge Queries. In the dialog, select Store_Locations as the second table and choose Store_ID as the join key in both tables. Set the Join Kind to Left.

Step 3 : Expand the merged column to include Store_Type, Region, and Opening_Year. Click Close & Apply.







```
= Table.ExpandTableColumn(#"Merged Queries", "Store_Locations", {"Store_ID", "Store_Type", "Region",
```

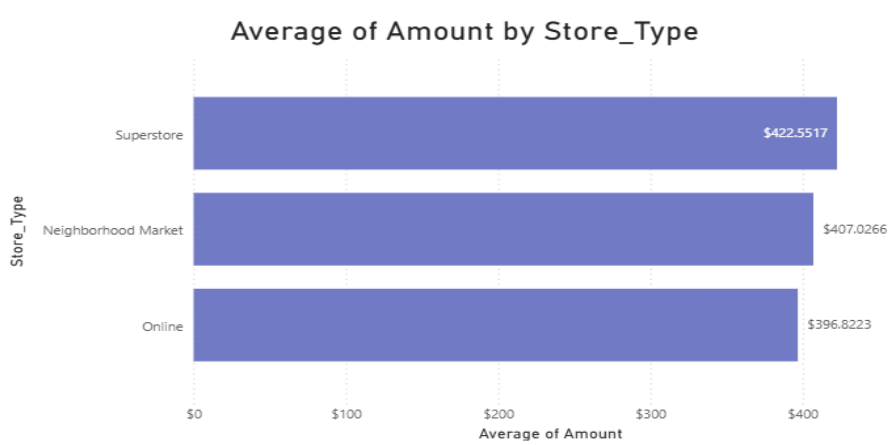
	Store_Locations.Store_ID	Store_Locations.Store_Type	Store_Locations.Region	Store_Locations.Opening_Year
1	S17	Neighborhood Market	South	
2	S17	Neighborhood Market	South	
3	S01	Online	East	
4	S06	Online	South	
5	S06	Online	South	
6	S08	Neighborhood Market	North	
7	S08	Neighborhood Market	North	
8	S03	Online	West	
9	S03	Online	West	
10	S03	Online	West	
11	S19	Online	Central	
12	S19	Online	Central	
13	S04	Online	Central	
14	S05	Online	Central	
15	S05	Online	Central	
16	S14	Neighborhood Market	Central	
17	S13	Superstore	West	
18	S13	Superstore	West	
19	S16	Neighborhood Market	South	
20	S12	Neighborhood Market	East	
21	S20	Neighborhood Market	West	
22	S03	Online	West	
23	S09	Superstore	Central	
24	S14	Neighborhood Market	Central	
25	S03	Online	West	
26	S17	Neighborhood Market	South	

Task 5.2: Visualize

A) Avg. transaction amount by Store Type

Visual: Clustered bar Chart

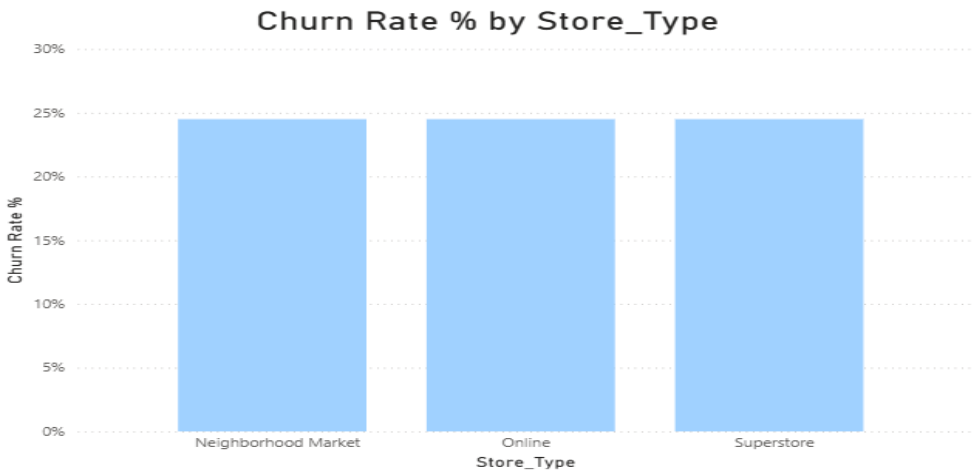
Field: store_type->axis , average amount->values



B) Churn rate by store type

Visual: Clustered column Chart

Field: store_type->axis , churn rate%->values

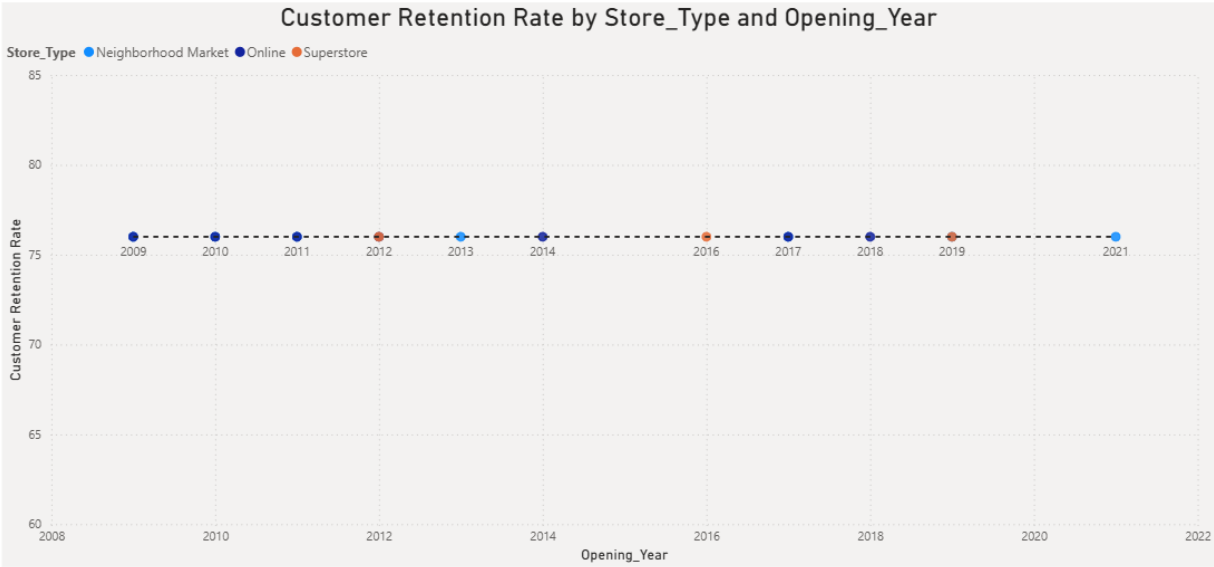


C) Correlation between store opening year & retention

Visual: Scatter Chart

Field: opening_year->x-axis , retention rate%-> y axis,

store_type-> legend



TASK6: Customer Lifetime Value (CLV) Analysis

Task 6.1: CLV = Total Amount Spent / Membership Duration (Years)

Step 1: create total amount spend column

Total Amount Spent =
SUMX(RELATEDTABLE(Customer_Transactions),
Customer_Transactions[Amount])

Step 2: Create CLV Column

CLV
=DIVIDE(CALCULATE(SUM(Customer_Transactions[Amount])),

MAX(Customer_Demographics[Membership_Duration_Years]),0)

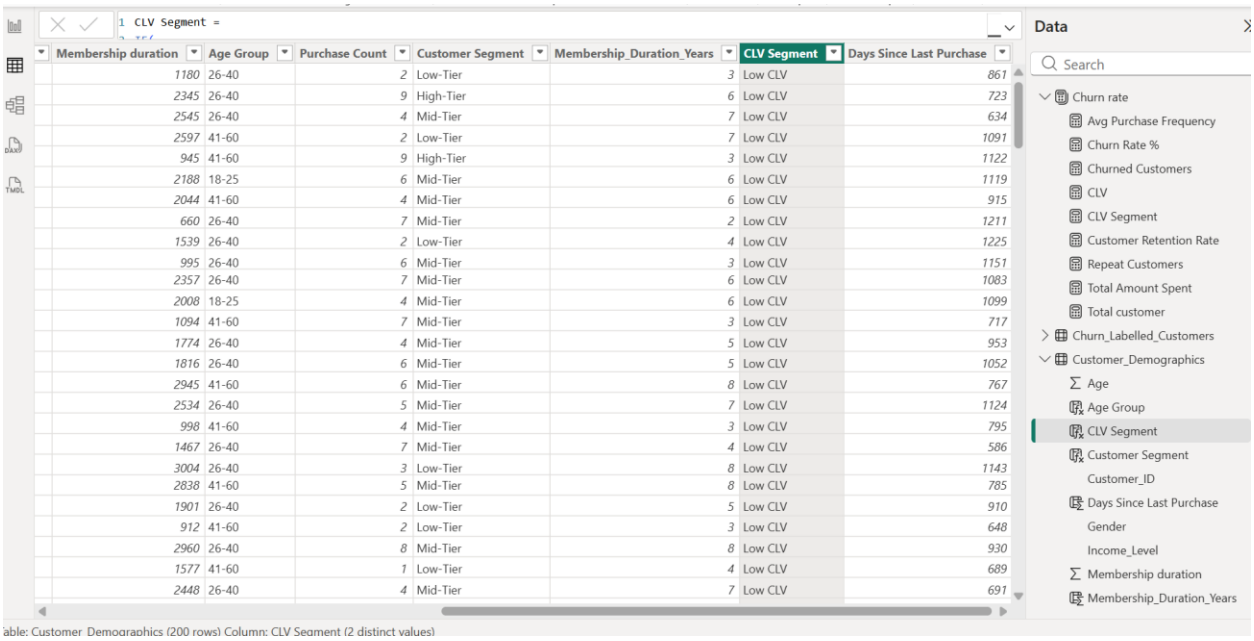
Task 6.2: Segment customers into Low, High CLV

○ Above Average CLV as High

○ Below Average CLV value- Low

CLV Segment =

IF(CALCULATE(SUM(Customer_Transactions[Amount])) / MAX(Customer_Demographics[Membership_Duration_Years]) >= 515.67,"High CLV", "Low CLV")



Membership duration	Age Group	Purchase Count	Customer Segment	Membership_Duration_Years	CLV Segment	Days Since Last Purchase
1180	26-40	2	Low-Tier	3	Low CLV	861
2345	26-40	9	High-Tier	6	Low CLV	723
2545	26-40	4	Mid-Tier	7	Low CLV	634
2597	41-60	2	Low-Tier	7	Low CLV	1091
945	41-60	9	High-Tier	3	Low CLV	1122
2188	18-25	6	Mid-Tier	6	Low CLV	1119
2044	41-60	4	Mid-Tier	6	Low CLV	915
660	26-40	7	Mid-Tier	2	Low CLV	1211
1539	26-40	2	Low-Tier	4	Low CLV	1225
995	26-40	6	Mid-Tier	3	Low CLV	1151
2357	26-40	7	Mid-Tier	6	Low CLV	1083
2008	18-25	4	Mid-Tier	6	Low CLV	1099
1094	41-60	7	Mid-Tier	3	Low CLV	717
1774	26-40	4	Mid-Tier	5	Low CLV	953
1816	26-40	6	Mid-Tier	5	Low CLV	1052
2945	41-60	6	Mid-Tier	8	Low CLV	767
2534	26-40	5	Mid-Tier	7	Low CLV	1124
998	41-60	4	Mid-Tier	3	Low CLV	795
1467	26-40	7	Mid-Tier	4	Low CLV	586
3004	26-40	3	Low-Tier	8	Low CLV	1143
2838	41-60	5	Mid-Tier	8	Low CLV	785
1901	26-40	2	Low-Tier	5	Low CLV	910
912	41-60	2	Low-Tier	3	Low CLV	648
2960	26-40	8	Mid-Tier	8	Low CLV	930
1577	41-60	1	Low-Tier	4	Low CLV	689
2448	26-40	4	Mid-Tier	7	Low CLV	691

Task 6.3: Visualize:

A) CLV vs Days Since Last Purchase:

Create Day since last purchase column

Days Since Last Purchase = DATEDIFF(RELATED

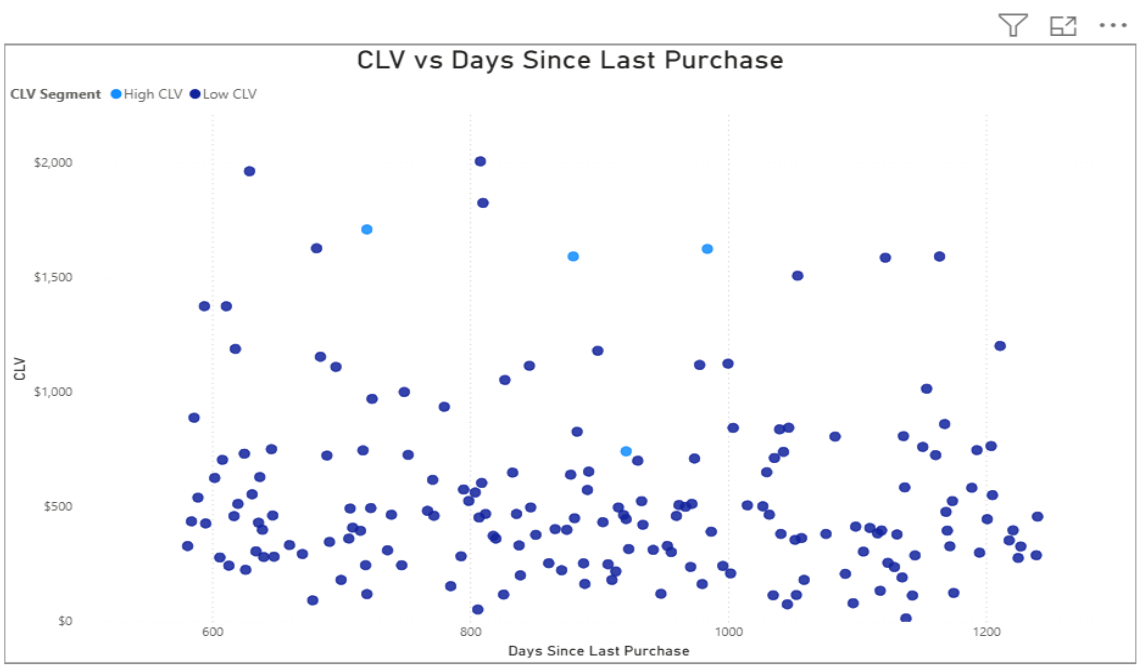
(Churn_Labelled_Customers[Last_Purchase_Date]), TODAY(), DAY)

```
Days Since Last Purchase = DATEDIFF(RELATED
(Churn_Labelled_Customers[Last_Purchase_Date]), TODAY(),  DAY)
```

Membership duration	Age Group	Purchase Count	Customer Segment	Membership_Duration_Years	CLV Segment	Days Since Last Purchase
1180	26-40	2	Low-Tier	3	Low CLV	861
2345	26-40	9	High-Tier	6	Low CLV	723
2545	26-40	4	Mid-Tier	7	Low CLV	634
2597	41-60	2	Low-Tier	7	Low CLV	1091
945	41-60	9	High-Tier	3	Low CLV	1122
2188	18-25	6	Mid-Tier	6	Low CLV	1119
2044	41-60	4	Mid-Tier	6	Low CLV	915
660	26-40	7	Mid-Tier	2	Low CLV	1211
1539	26-40	2	Low-Tier	4	Low CLV	1225
995	26-40	6	Mid-Tier	3	Low CLV	1151
2357	26-40	7	Mid-Tier	6	Low CLV	1083
2008	18-25	4	Mid-Tier	6	Low CLV	1099
1094	41-60	7	Mid-Tier	3	Low CLV	717
1774	26-40	4	Mid-Tier	5	Low CLV	953
1816	26-40	6	Mid-Tier	5	Low CLV	1052
2945	41-60	6	Mid-Tier	8	Low CLV	767
2534	26-40	5	Mid-Tier	7	Low CLV	1124
998	41-60	4	Mid-Tier	3	Low CLV	795
1467	26-40	7	Mid-Tier	4	Low CLV	586
3004	26-40	3	Low-Tier	8	Low CLV	1143
2838	41-60	5	Mid-Tier	8	Low CLV	785
1901	26-40	2	Low-Tier	5	Low CLV	910
912	41-60	2	Low-Tier	3	Low CLV	648
2960	26-40	8	Mid-Tier	8	Low CLV	930
1577	41-60	1	Low-Tier	4	Low CLV	689
2448	26-40	4	Mid-Tier	7	Low CLV	691

Visual: Scatter Chart

Field: Days since last purchase ->x axis, sum of CLV->y axis, CLV segment->legend



Interpretation: The chart visualizes a critical risk area. High-CLV customers in the top-right quadrant have high lifetime value but haven't shopped in 90 days, highlighting high-value customers at immediate risk of churning.

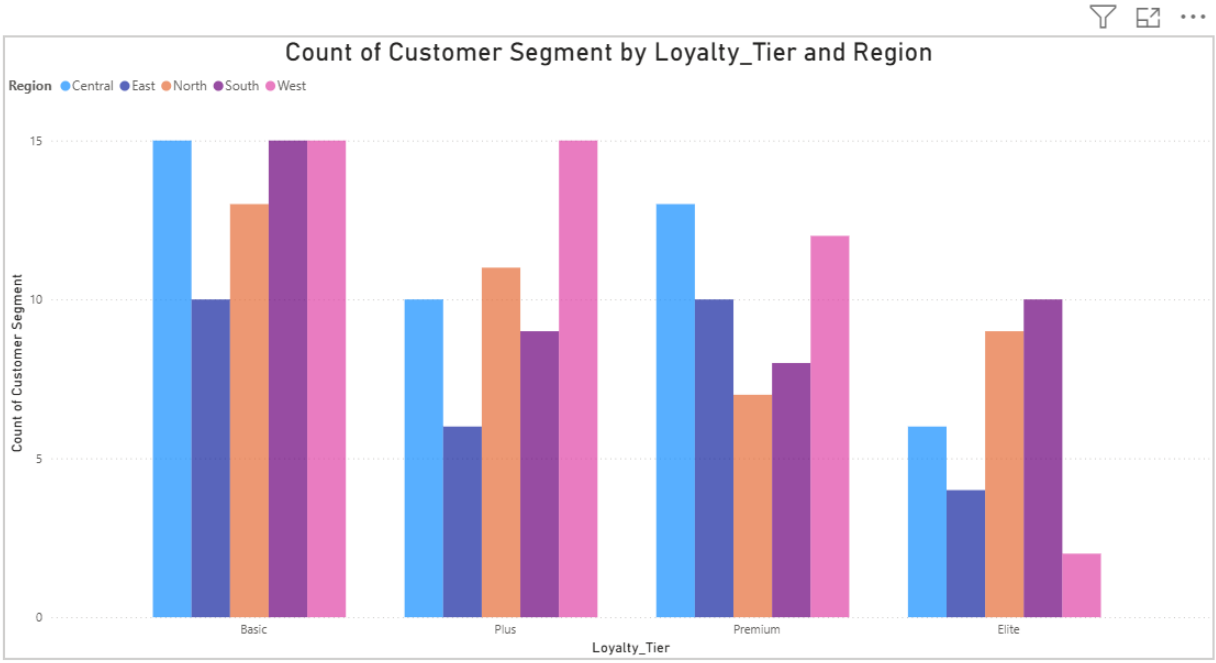
B) CLV by Loyalty Tier & Region

Insert a Clustered Column Chart.

X-Axis: Loyalty_Program[Loyalty_Tier].

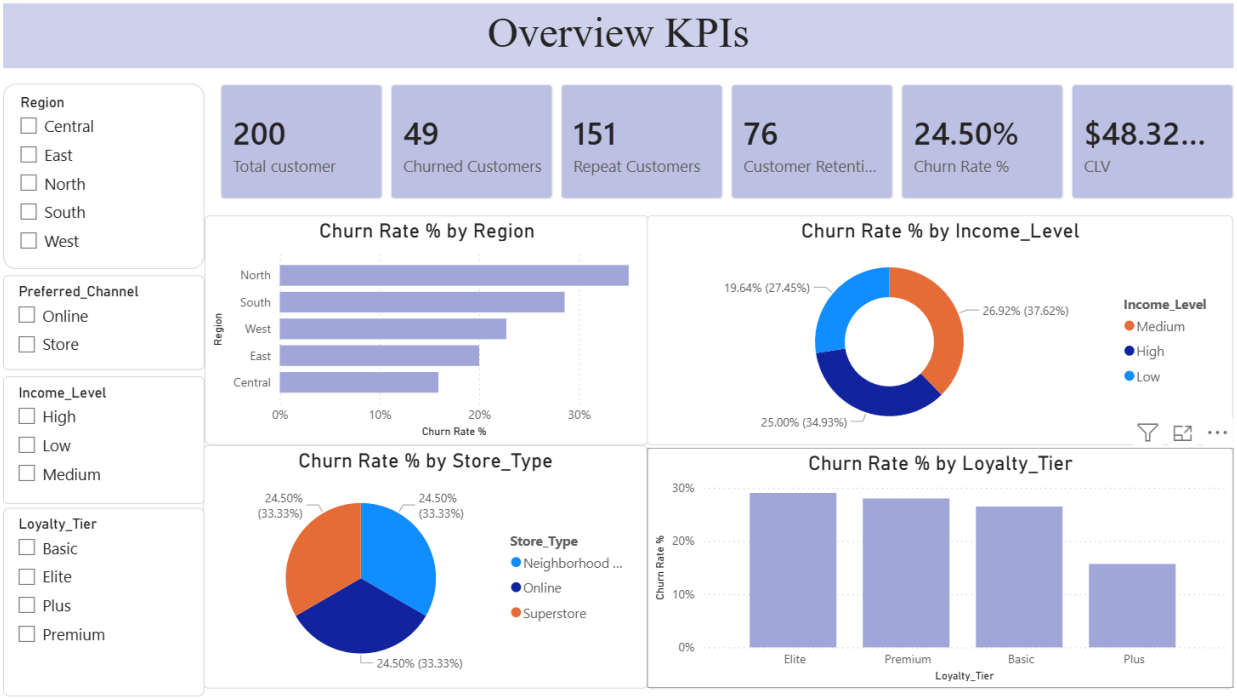
Y-Axis: CLV Segment.

Legent : Region



Task 7: Final Dashboard & Executive Summary

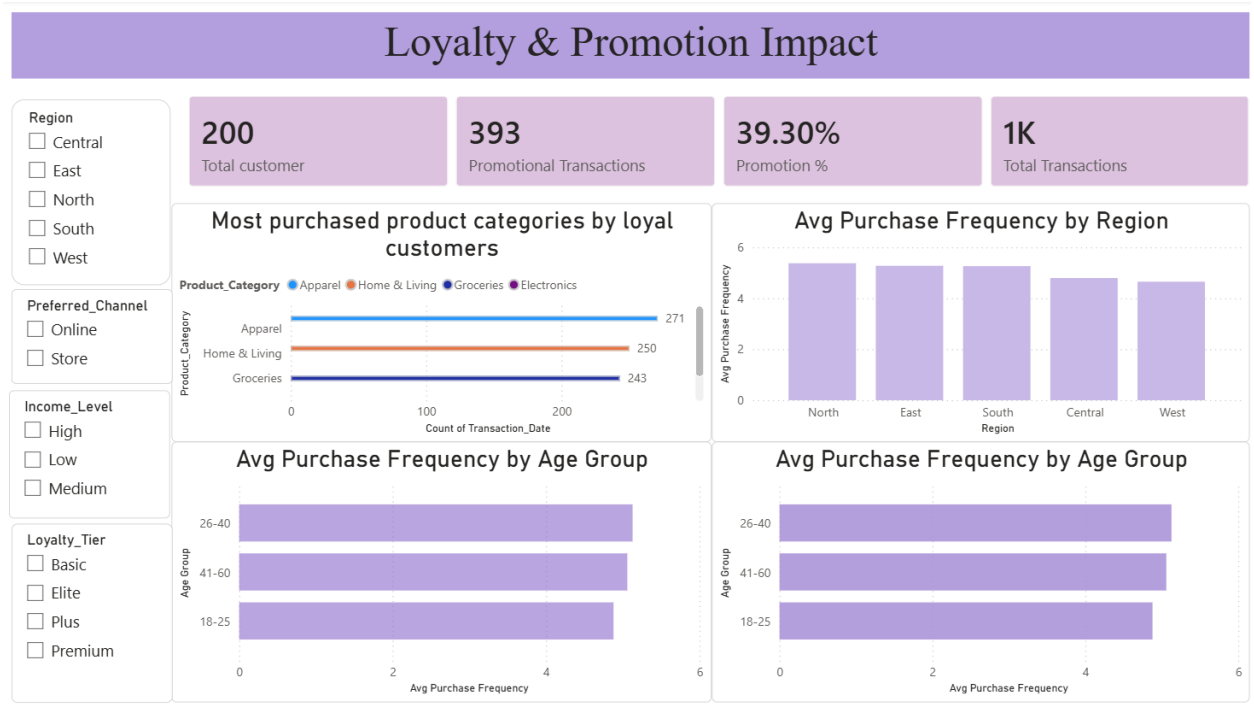
DASHBOARD:1: KPIs (Churn, CLV, Repeat Rate)



Key Demographic Findings:

- Geographic Risk: Churn peaks in the North region (29.0%) and remains lowest in the Central region (15.9%).
- Channel Vulnerability: In-Store shoppers churn at a higher rate (26.9%) compared to Online platform shoppers (22.1%).
- Socioeconomic Segment: Medium-Income customer accounts represent the highest individual share of churned users (24.9%).

DASHBOARD 2: Loyalty & Promotion Impact

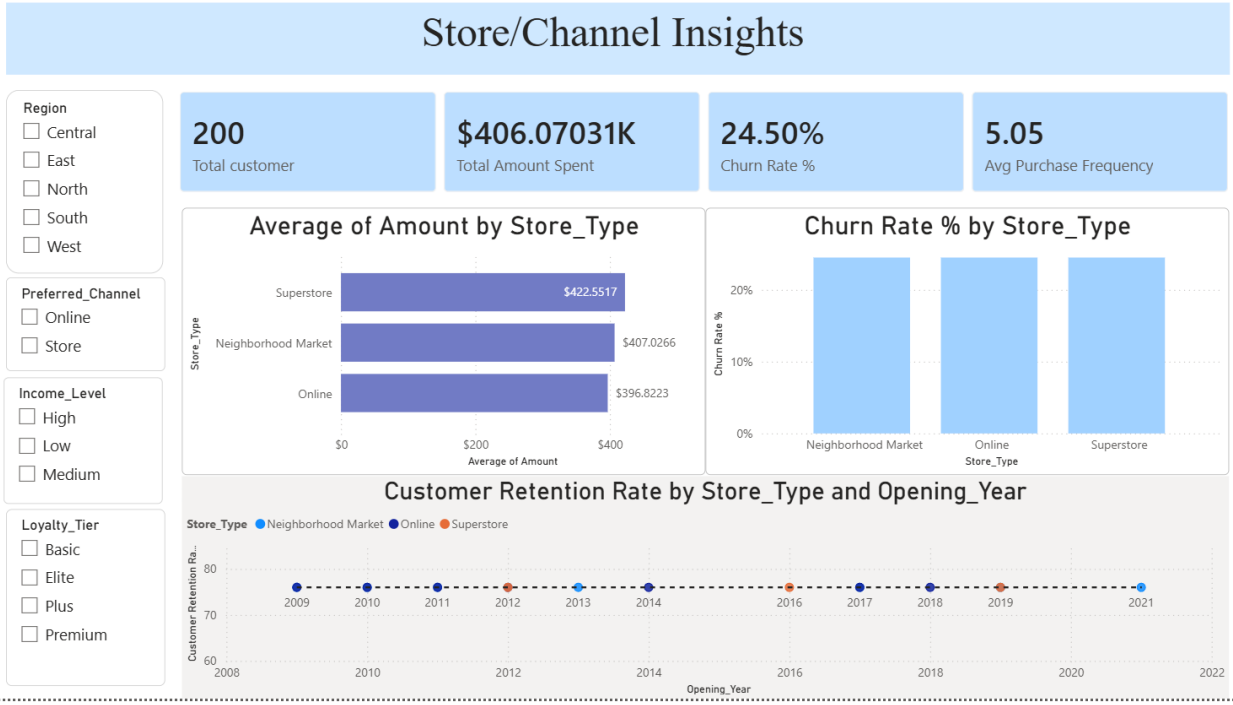


- Active Promotional Share: 39.30% of total transactions (393 out of 1,000) had marketing promotions applied.

The Promotion Spend Paradox:

- Average Spend without Promotion: 411
- Average Spend with Promotion: 398
- The Impact: Promotions are actually diluting average basket sizes by 13 per visit. Discounts are primarily being applied to items customers would have purchased anyway, rather than successfully encouraging shoppers to add extra, high-value items to their carts.
- **The Attrition Link**: Basic loyalty tiers exhibit healthy point-to-redemption activity (154k earned vs. 122k redeemed).

DASHBOARD 3: Store/Channel Insights

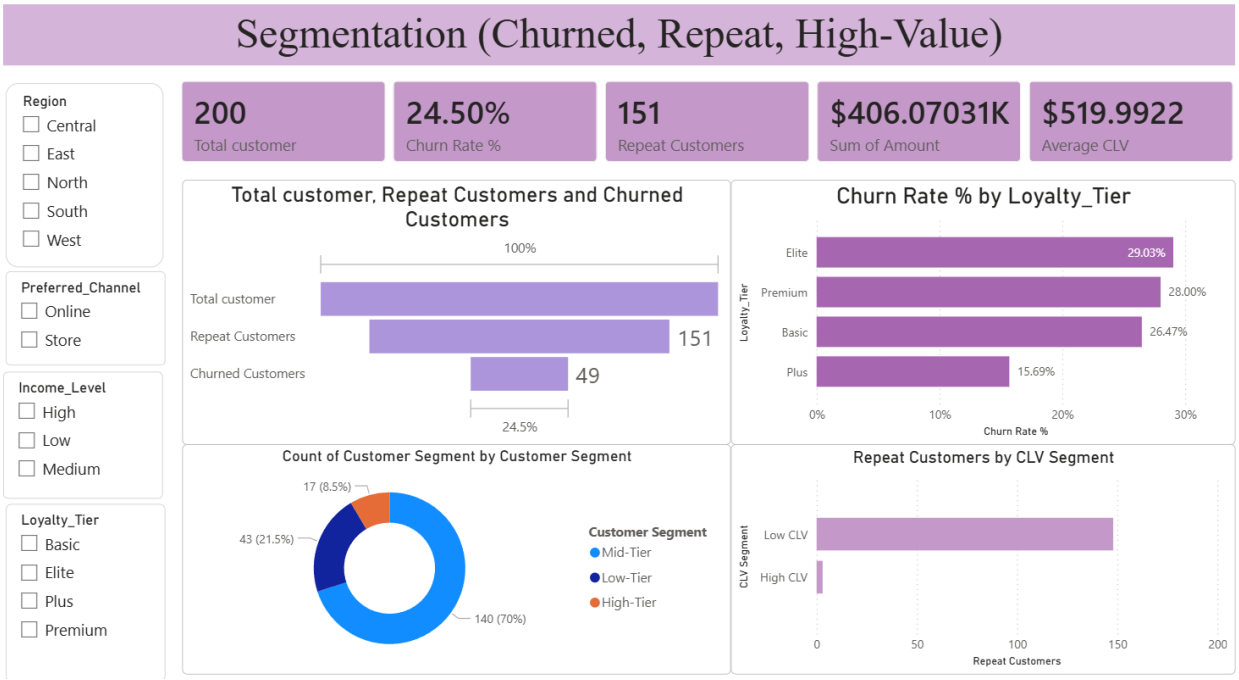


- Format Spend: Supercenters lead in transactional value (averaging $423 per cart), followed by Neighborhood Markets ($407) and Online Channels ($397).
- Format Churn: Neighborhood Markets exhibit the highest operational risk, with store-level churn peaking at $25.7%.

Store Age vs. Retention Correlation:

- Legacy Stores (Opened pre-2012): Show low customer retention (retaining only 68% to 73%).
- Modern Stores (Opened 2018–2020): Achieve superior customer retention rates (up to 94%).
- Takeaway: Target's newer "smart-layouts" (featuring dedicated online order pickup zones, drive-up parking, and localized inventory) are significantly more effective at retaining customers than legacy store layout

DASHBOARD 4: Segmentation (Churned, Repeat, High-Value)



- Average Baseline CLV:$509.35 annually per customer.
- High-Value VIPs Count: 68customers classified as High CLV (above-average value).
- High-Value Churn Rate: 28.50%$of our High-CLV base is churning, posing a significant revenue risk.

Slicers: Region, Channel, Income, Loyalty Tier:

To ensure a seamless user experience, we enabled Power BI's Sync Slicers feature on our four global slicers (Region, Preferred_Channel, Income_Level, and Loyalty_Tier). This synchronization links Page 1, Page 2, Page 3, and Page 4 together. When an executive selects a specific demographic segment on the landing page, the entire report automatically filters in the background, keeping the analysis unified as they navigate through the pages.

Executive Summary:

Provide Top 3 Recommendations for Target:

1) Which customers to prioritise for retention?

Focus immediately on High CLV customers in the Basic or Plus loyalty tier who have not purchased in 60+ days. These are high-value, at-risk customers. Deploy personalised win-back emails with an exclusive offer (e.g., 500 bonus points or 15% off their most-purchased category) before the 90-day churn threshold is reached.

2) Which channels are underperforming?

If the Online channel shows higher churn than Store, invest in the post-purchase digital experience: send order confirmation with a loyalty points summary, add a personalised 'You Might Also Like' recommendation email 7 days after purchase, and offer online-exclusive bonus points events to increase stickiness.

3) How to strengthen loyalty engagement?

The gap between Points Earned and Points Redeemed signals that customers do not perceive the redemption process as easy or rewarding. Launch a 'Points Expiry Alert' 30 days before expiration, introduce a 2× redemption-value event quarterly, and simplify the redemption flow in the app to no more than two taps.